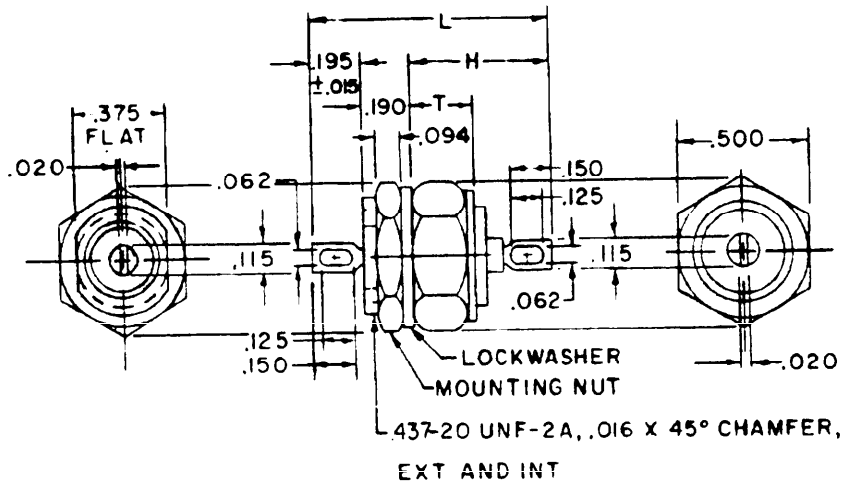
MIL-C-10950/7C
12 June 1978
SUPERSEDING
MIL-C-10950/7B
8 July 1968

MILITARY SPECIFICATION SHEET

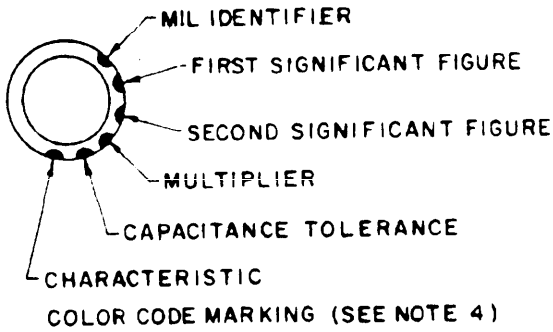
CAPACITORS, FIXED, MICA DIELECTRIC, BUTTON STYLE (FEEDTHRU, HERMETIC SEAL) STYLES CB60, CB61, AND CB62

This specification is approved for use by all Departments and Agencies of the Department of Defense.

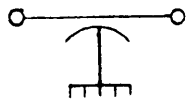
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-10950.



INCHES	MM
.015	.38
.016	.41
.020	.51
.062	1.57
.094	2.39
.115	2.92
.125	3.18
.150	3.81
.190	4.83
.195	4.95
.375	9.52
.437	11.10
.500	12.70



Style	Dimensions T±.015(.38)	L (Ref)	H±.030(.76)
CB60	.125 (3.18)	.790 (20.07)	.405 (10.29)
CB61	.150 (3.81)	.815 (20.70)	.430 (10.92)
CB62	.230 (5.84)	.895 (22.73)	.510 (12.95)



CIRCUIT DIAGRAM

NOTES:
1. All dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is .xxx ±.010 (.25 mm).
4. Color-coded section shall not exceed 240°.
5. Ferrous material may be used.

FIGURE 1. Capacitors, (feedthru, hermetic seal), styles CB60, CB61, and CB62.

Ⓒ denotes changes

FSC 5910

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© TABLE I. Styles CB60, CB61 and CB62.

Part number 1/	Capacitance	DC rated voltage	Characteristic available at capacitance tolerance				Terminal assemblies
			+1% (F)	+2% (G)	+5% (J)	+10% (K)	
	pF	Volts					
CB60R-050K	5	500				B	R
CB60R-100K	10	500				D	R
CB60R-150K	15	500				D	R
CB60R-200J	20	500			D	-	R
CB60R-220-	22	500			D	D	R
CB60R-330-	33	500			D	D	R
CB60R-470-	47	500			D	D	R
CB60R-680-	68	500	D,F	D,F	D,F	D	R
CB60R-101-	100	500	D,F	D,F	D,F	D	R
CB60R-151-	150	500	D,F	D,F	D,F	D	R
CB60R-221-	220	500	D,F	D,F	D,F	D	R
CB60R-331-	330	500	D,F	D,F	D,F	D	R
CB60R-471-	470	500	D,F	D,F	D,F	D	R
CB61R-681-	680	500	D,F	D,F	D,F	D	R
CB61R-821-	820	500	D,F	D,F	D,F	D	R
CB61R-102-	1,000	500	D,F	D,F	D,F	D	R
CB62R-122-	1,200	500	D,F	D,F	D,F	D	R
CB62R-152-	1,500	500	D,F	D,F	D,F	D	R
CB62R-182-	1,800	500	D,F	D,F	D,F	D	R
CB62R-222-	2,200	500	D,F	D,F	D,F	D	R
CB62R-242-	2,400	500	D,F	D,F	D,F	D	R

1/ Complete part number will include additional symbols to indicate characteristic and capacitance tolerance.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Case: See figure 1.
Terminals: See figure 1.
Capacitance value: See table I.
Capacitance tolerance: See table I.
DC rated voltage: 500 volts.
Dissipation factor: See figure 1 of MIL-C-10950.
Characteristics; See table I.

©

Symbol	Temperature coefficient	Capacitance drift
D	-100 to +100	0.3 percent or 0.3 pF whichever is greater.
F	-60 to +60	±(0.1 percent ± 0.1 pF).

Operating temperature range: -55° to +150°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Barometric pressure: Method 105 of MIL-STD-202, condition C (70,000 ft.)
Test potential - 125 percent of rated dc voltage for not less than 1 second nor more than 5 seconds. Surge current shall not exceed 5 mA.

Insulation resistance: Method 302 of MIL-STD-202, condition A or B.
At 25°C - 100,000 megohms minimum.
At 150°C - 15,000 megohms minimum.

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Capacitance: Method 305 of MIL-STD-202.
Measurement frequency:
Capacitance values of 1000 pF or less - 1 MHz ±100 kHz.
Capacitance values of greater than 1000 pF - 1 kHz ±100 Hz.

Shock medium impact: Method 205 of MIL-STD-202, condition C (50 G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition B.

Moisture resistance: Test method 106 of MIL-STD-202.
Insulation resistance: Not less than 100,000 megohms.
Capacitance change: Within 1 percent of the initial value or 0.5 pF, whichever is greater.
Dissipation factor - Initial requirement.

(C) Thermal shock and immersion:
Thermal shock: Test method 107 of MIL-STD-202, condition A, except temperature for step 3 shall be $150^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}$ C.

Immersion: Test method 104, of MIL-STD-202, condition A.
Insulation resistance: Not less than 100,000 megohms.
Capacitance change: Within 1 percent of the initial value or 0.5 pF, whichever is greater.

Life: 2000 ±12 hours at 150° C $^{+3^{\circ}}_{-0^{\circ}}$ C.
Insulation resistance: Greater than 100,000 megohms.
Capacitance change: Within 1 percent of the initial value or 0.5 pF, whichever is greater.

Temperature coefficient and capacitance drift: Within the limits specified for the applicable characteristic.

Torque: In accordance with MIL-C-10950.
Mounting - normal mounting means.
Load - 4 ounce-inches.

NOTES:
1. Conversion of dash numbers to part numbers is as shown in table II.
(C)2. Characteristic E has been replaced by characteristic F.

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TABLE II. Dash number to part number conversion.

Dash number in MIL-C-10950/7A		Converted to part number 1/ in MIL-C-10950/7B and 7C
Characteristic D	Characteristic E	
0001	0002	CB60R-680F
0003	0004	CB60R-101F
0005	0006	CB60R-151F
0007	0008	CB60R-221F
0009	0010	CB60R-331F
0011	0012	CB60R-471F
0013	0014	CB61R-681F
0015	0016	CB61R-821F
0017	0018	CB61R-102F
0019	0020	CB62R-122F
0021	0022	CB62R-152F
0023	0024	CB62R-182F
0025	0026	CB62R-222F
0027	0028	CB62R-242F
0029	0030	CB60R-680G
0031	0032	CB60R-101G
0033	0034	CB60R-151G
0035	0036	CB60R-221G
0037	0038	CB60R-331G
0039	0040	CB60R-471G
0041	0042	CB61R-681G
0043	0044	CB61R-821G
0045	0046	CB61R-102G
0047	0048	CB62R-122G
0049	0050	CB62R-152G
0051	0052	CB62R-182G
0053	0054	CB62R-222G
0055	0056	CB62R-242G
0057		CB60RD220J
0058		CB60RD330J
0059		CB60RD470J
0060	0061	CB60R-680J
0062	0063	CB60R-101J
0064	0065	CB60R-151J
0066	0067	CB60R-221J
0068	0069	CB60R-331J
0070	0071	CB60R-471J
0072	0073	CB61R-681J
0074	0075	CB61R-821J
0076	0077	CB61R-102J

1/ See footnote at end of table.

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TABLE II. Dash number to part number conversion (continued).

Dash number in MIL-C-10950/7A		Converted to part number 1/ in
Characteristic D	Characteristic E	MIL-C-10950/7B and 7C
0078	0079	CB62R-122J
0080	0081	CB62R-152J
0082	0083	CB62R-182J
0084	0085	CB62R-222J
0086	0087	CB62R-242J
0088		CB60RD150K
0089		CB60RD220K
0090		CB60RD330K
0091		CB60RD470K
0092		CB60RD680K
0093		CB60RD101K
0094		CB60RD151K
0095		CB60RD221K
0096		CB60RD331K
0097		CB60RD471K
0098		CB61RD681K
0099		CB61RD821K
0100		CB61RD102K
0101		CB62RD122K
0102		CB62RD152K
0103		CB62RD182K
0104		CB62RD222K
0105		CB62RD242K

1/ The complete part number shall include symbols for terminal assembly and characteristic, as applicable.

Custodians:
Army - EL
Navy - EC
Air Force -11

Review activities:
Army - AR
Navy - SH
Air.Force - 17, 85
DLA - ES

User activities:
Navy - AS, CG, OS

Preparing activity:
Navy - EC

Agent:
DLA - ES

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MIL-C-10452/7C 12 JUNE 1978

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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